



Strongly Connected Components and Topological Sort.

1 Topological Sort

Topological sort of a directed graph is a ordering of its vertices such that for every directed edge (u, v) from vertex u to vertex v , u comes before v in the ordering. The native application is in determining a compilation order. Imagine a huge source code, a compiler has to compile in such an order that when compiling a function, all the functions on which it depends have already been compiled.

Definition 1. (connectivity)

1. Let $G = (V, E)$ be an undirected graph. A **connected component** of G is a subset $U \subseteq V$ in which there exists a path between every two vertices, and U is 'maximal'¹ in the sense that any extension of it by a vertex results in two vertices that are not connected.
2. A non-directed graph G is said to be a **connected** graph if it only has one connected component.
3. Let $G = (V, E)$ be a directed graph. A **strongly connected component** of G is a subset $U \subseteq V$ in which for any pair of vertices $u, v \in U$, there exist both a directed path from u to v and a directed path from v to u , and U is also 'maximal' in the sense that any extension of it by a vertex results in two vertices that are not connected (from both sides).
4. A directed graph G is said to be a **strongly connected** if it only has one strongly connected component.
5. A **directed acyclic graph (DAG)** is a directed graph with no cycles.

1.1 Depth First Search (DFS)

As its name implies, depth-first search searches "deeper" in the graph whenever possible. Depth-first search explores edges out of the most recently discovered vertex v that still has unexplored edges leaving it. Once all of v 's edges have

¹Local maximum.



Figure 5.1: Topological sort - the numbers x/y denote the entering and the exiting times to and from the vertex (pre and post).

been explored, the search "backtracks" to explore edges leaving the vertex from which v was discovered. This process continues until all vertices that are reachable from the original source vertex have been discovered. If any undiscovered vertices remain, then depth-first search selects one of them as a new source, repeating the search from that source. The algorithm repeats this entire process until it has discovered every vertex.

<pre> 1 DFS(G): 2 for $v \in V$ do 3 $v.visited \leftarrow False$ 4 end 5 $time \leftarrow 1$ 6 for $v \in V$ do 7 if <i>not</i> $v.visited$ then 8 $\pi(v) \leftarrow null$ 9 Explore(G, v) 10 end 11 end 1 Explore(G, v): 2 Previsit(v) 3 for $(v, u) \in E$ do 4 if <i>not</i> $u.visited$ then 5 $\pi(u) \leftarrow v$ 6 Explore(G, u) 7 end 8 end 9 Postvisit(v) </pre>	<pre> 1 Previsit(v): 2 $pre(v) \leftarrow time$ 3 $time \leftarrow time + 1$ 1 Postvisit(v): 2 $post(v) \leftarrow time$ 3 $time \leftarrow time + 1$ </pre>
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Properties of depth-first search. Depth-first search yields valuable information about the structure of a graph. Perhaps the most basic property of depth-first

search is that the predecessor subgraph G_π does indeed form a forest of trees since the structure of the depth-first trees exactly mirrors the structure of recursive calls of explore-function. That is, $u = \pi(v)$ if and only if $\text{explore}(G, v)$ was called during a search of u 's adjacency list.

Additionally, vertex v is a descendant of vertex u in the depth-first forest if and only if v is discovered during the time in which $\text{pre}(u)$ has already been set but $\text{post}(u)$ hasn't². Another important property of depth-first search is that discovery and finish times have a parenthesis structure. If the explore procedure were to print a left parenthesis " $(u$ " when it discovers vertex u and to print a right parenthesis " $)u$ " when it finishes u , then the printed expression would be well-formed in the sense that the parentheses are properly nested.

The following theorem provides another way to characterize the parenthesis structure.

Theorem 1 (Parenthesis theorem). *In any depth-first search of a (directed or undirected) graph $G = (V, E)$, for any two vertices u and v , exactly one of the following three conditions holds:*

1. *the intervals $[\text{pre}(u), \text{post}(u)]$ and $[\text{pre}(v), \text{post}(v)]$ are entirely disjoint, and neither u nor v is a descendant of the other in the depth-first forest.*
2. *the interval $[\text{pre}(u), \text{post}(u)]$ is contained entirely within the interval $[\text{pre}(v), \text{post}(v)]$, and u is a descendant of v in a depth-first tree, or*
3. *the interval $[\text{pre}(v), \text{post}(v)]$ is contained entirely within the interval $[\text{pre}(u), \text{post}(u)]$, and v is a descendant of u in a depth-first tree.*

Let us use the following notation: we say that line (x) of the recursive call 'Explore(G, v)' is executed at time t if at time t the running of the algorithm got to line number x of Explore where the vertex argument is v . Notice that since Explore is called **only once** on each vertex, the notation is well defined.

Proof. It's enough to prove that for any pair of vertices v, u , the inequality $\text{pre}(v) < \text{pre}(u) < \text{post}(v)$ implies $\text{pre}(v) < \text{pre}(u) < \text{post}(u) < \text{post}(v)$.

Consider the call 'Explore(G, v)'. Since the algorithm sets the value of $\text{post}(v)$ at line number (9), and it makes all of its recursive calls between lines (3)-(8), we have that the algorithm returns from its recursive calls before time $\text{post}(v)$. Furthermore, any line that has been executed during times $[\text{pre}(v), \text{post}(v)]$ is either a line of the call 'Explore(G, v)' or a line of one of the recursive calls.

In particular, we have that the line which sets $\text{pre}(u)$ is executed during times $[\text{pre}(v), \text{post}(v)]$, and therefore as part of a recursive call. In addition, because the recursive call was returned during the time interval, the assignment of $\text{post}(u)$ is made before $\text{post}(v)$. Thus, we have both $\text{pre}(v) < \text{pre}(u) < \text{post}(u) < \text{post}(v)$ and that u is a descendant of v in π ³.

□

²In terms of the presentation from last week, when u 's color is gray

³The algorithm defines π to be the recursive tree

Corollary 1. Vertex v is a proper descendant of vertex u in the depth-first forest for a (directed or undirected) graph G if and only if $\text{pre}(u) < \text{pre}(v) < \text{post}(v) < \text{post}(u)$.

2 DFS on DAGS

We are about to come with an algorithm for sorting a DAG's vertices according to their order. The following two claims clarify the roles of pre and post in determining such order.

Claim 1. Let $G = (V, E)$ be a DAG, and let $v, u \in V$ such $u \rightsquigarrow v$ (There is a path from u to v in G). And let π be a forest returned from the call **DFS**(G), If $\text{pre}(u) < \text{pre}(v)$ then v is a descendant of u in π .

To prove Claim 1, we will define a graph G' that satisfies:

1. There is only single path between u to v in G' .
2. Let π' be the depth forest returned by the running of DFS over G' , then if v is a descendant of u in π' then is also a descendant of u in π .

Definition 2. Consider a graph G and vertices u, v such that there is a path from u to v . Let $e_1, e_2, \dots, e_m$ be the edges of the graph ordered by the access time of the DFS running over G . Let $G' \subset G$ be defined by taking the minimal prefix $e_1, e_2, \dots, e_i$ such that there is still a path from u to v . Let π' be the forest returned by the DFS running over G' , and notice that $\pi' \subset \pi$. Observe that by minimality, there is only one path between u and v in G' .

Proof. Consider vertices v, u such $u \rightsquigarrow v$, in G , $\text{pre}(u) < \text{pre}(v)$ and G' as defined above. Assume without loss of generality that v is the first of such vertex on the path that is not a descendant of u in π' . Let v' be predecessor to v on that path, namely the last vertex on the ' $u \rightsquigarrow v$ ' path before v .

Either $\text{pre}(v') < \text{pre}(u)$ or v' is a descendant of u . In the first case, we get an immediate contradiction to $\text{pre}(v) < \text{pre}(u)$. In the second case, we consider the recursive call '**Explore**(G', v')', and we have that $\text{pre}(v) < \text{pre}(v')$, or otherwise the algorithm will set $\pi'(v) \leftarrow v'$, and hence v is a descendant of u in π' .

So, let's focus on the case where $\text{pre}(v) < \text{pre}(v')$. Let v'' be the predecessor of v' . Notice that since there is a single path from u to v , we have that $\text{pre}(v) < \text{pre}(v')$ implies $\text{pre}(v) < \text{pre}(v'')$.⁴ Then we continue by induction until we get $\text{pre}(v) < \text{pre}(u)$, which is a contradiction. □

Claim 2. Let $G = (V, E)$ be a DAG, and let $v, u \in V$ such $u \rightsquigarrow v$ (There is a path from u to v in G). Then the call **DFS**(G) assigns $\text{post}(u) > \text{post}(v)$.

Proof. Suppose, toward a contradiction, that $\text{post}(u) < \text{post}(v)$, and split into cases:

⁴That is, upon calling '**Explore**(G', v'')', the algorithm will first walk through paths that do not reach v , and then call '**Explore**(G', v')'.

- Either $\text{pre}(v) < \text{pre}(u)$ thus, from the parenthesis theorem, $\text{pre}(v) < \text{pre}(u) < \text{post}(u) < \text{post}(v)$, and u is a descendant of v in π . Therefore, there is a path from v to u in G . Combining this with the given that there is a path from u to v , there is a cycle in G , which contradicts G being a DAG.
- Or, $\text{pre}(v) > \text{pre}(u)$, Thus, by Claim 1, we get that v is a descendant of u in π , and by combining the parenthesis theorem, we get the desired.

□

Claim 2 defines an algorithm for sorting the vertices of a DAG according to dependency, namely v appears before u only if there is not a directed path from u to v , which we provide in Algorithm 1. Notice that since the post values are integers bounded by $|E|$, the sorting can be done using counting sort, and the overall running time is $O(|V| + |E|)$.

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1 pre, post  $\leftarrow$  Call DFS( $G$ )
2 return  $G$ 's vertices in post-reversed ordered.

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Algorithm 1: TopologicalSort(G)

3 Strongly Connected Components

Denote by $G^\top$ the graph obtained by flipping the direction of each edge in G , namely, if $(u, v) \in G$, then $(v, u) \in G^\top$. Let's introduce the first version of an algorithm that extracts from a given graph G all its strongly connected components.

```

1 for  $v$  in  $V$  do
2    $\pi \leftarrow \text{Explore}(G, v)$ 
3    $\pi' \leftarrow \text{Explore}(G^\top, v)$ 
4   print  $\pi \cap \pi'$ 
5    $V \leftarrow V / \pi \cap \pi'$ 
6 end

```

Algorithm 2: Find SCCs slow version.

Correctness, Denote by C_w the subset of vertices that is printed at line (4) when the iteration is over w . Let u, v be vertices belonging to C_w . By the correctness of Explore, line (2) guarantees that $w \rightsquigarrow u, v$, and line (3) guarantees that $u, v \rightsquigarrow w$. Therefore, we get both $u \rightsquigarrow v$ and $v \rightsquigarrow u$. To complete the proof, one also has to show that the extension $C_w \cup \{z\}$ for any $z \in V \setminus C_w$ is not strongly connected. Showing that contradicts the correctness of Explore is left as an exercise.

Running Time, Denote by $T(G)$ the running time of the algorithm when called on graph G , and denote by C_1 the first strongly connected component that is found. Then:

$$T(G) = \Theta(|V| + |E|) + T(G/C_1) = O(|E| \cdot |V|)$$

4 SCCs in Linear Time

Now we present Algorithm 3, an algorithm to find the strongly connected components in linear time.

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1 pre, post  $\leftarrow$  DFS( $G^\top$ )
2  $V \leftarrow$  sorts  $V$  in reverse order of post.
3  $\pi \leftarrow$  DFS( $G$ )
4 return  $\pi$ 

```

Algorithm 3: Find SCCs fast version.

Claim 3. Define the relation $v \sim u$ to hold if u and v are on the same strong connected component. Then:

1. $\sim$ is an equivalence relation.
2. Define the graph $G' = (V', E')$ such that V' are the equivalence class respecting to $\sim$ and $\{u, v\} \in E'$ if there is a directed path in G from a vertex in u into a vertex in v . Then G' is a DAG.

Corollary 2. For $v \in V$, denote by $[v]$ the delegate of v according to $\sim$. Running the topological sort on G returns the vertices of G according to the topological order of G' , namely that v appears before u only if there is no a directed path from $[u]$ to $[v]$ in G' . The proof is identical to the proof of Claim 2.

For SCCs C_1 and C_2 such that there is a directed path from C_1 to C_2 , we have that $\max_{v \in C_1} \text{post}(v) \geq \max_{u \in C_2} \text{post}(u)$. That result wasn't proved in the recitation. Proof idea: The vertex with the maximal post value in C_i is the first vertex to be visited in that connected component. Consider G' as defined above, such that the delegates of each component are the first vertices to be visited. Then we get the inequality by Claim 2.

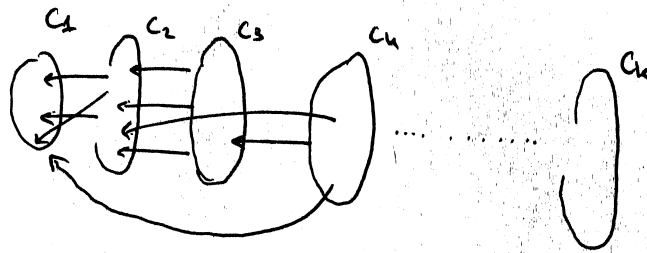


Figure 5.2: The vertices are 'ordered' according to the components' reversed order. One can imagine that at line (3), the algorithm calls 'Explore' first on the vertices in C_1 , then on the vertices in C_2 , and so on. Since there are no edges from C_i to C_j where $j > i$, it guarantees that the 'Explore' call will not get outside C_i . Hence, each tree in the returned π is associated with a single strongly connected component.

Correctness. Denote by $C_1, C_2, C_3, \dots, C_l$ the strongly connected components of G ordered by the topological sort of $G^\top$. We will prove by induction

that upon the i th calling to 'Explore' from the second DFS call (line (3)), the algorithm scans only the vertices at C_i .

1. Base. The first call to Explore is over the first vertex in the topological order of $G^\top$, denoted by v . That vertex belongs to C_1 , and from it, there is a directed path to any other vertex of C_1 , so it's left to show that no vertex outside C_1 is scanned on the first call. If there were $u \in C_j$ for $j > 1$ that was scanned, then it means that the algorithm crossed through an edge of the form (v', u') such that $v' \in C_1$ and $u' \in C_j$. But if $(v', u') \in E$, then $(u', v') \in E^\top$.

Thus, there is a directed path from u' to v in $G^\top$. Using Corollary 2, it holds that either $u' \in C_1$, which contradicts $j > 1$, or that $C_j \rightsquigarrow C_1$ and therefore $\max_{u \in C_j} \text{post}(u) > \max_{v \in C_1} \text{post}(v)$ (regarding $G^\top$), So the first vertex of C_j has to appear before v , which contradicts v being the first vertex in the order.

2. Assumption. Assume correction for any $i' < i$.
3. Step. Left as an exercise.